

PROGRAMME OF
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IMPLEMENTED BY



Info

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> CDSAPI setup

CDSAPI setup

The **Climate Data Store (CDS) Application Program Interface (API)** is a service providing programmatic access to CDS, ADS and CEMS Early Warning DS data. In this page you will find explanations and examples showing how to use the CDS API.

For troubleshooting, check [Common Error Messages for CDS Requests](#)

For advanced capabilities, check the [New API package for advanced users](#) section.



Windows users

Please read [How to install and use CDS API on Windows](#)





Please read [How to install and use CDS API on macOS](#)



Linux users

Please proceed as follows:

1. Setup the CDS Api key

Here is how to setup the CDS Api key:

1. If you do not have an account yet, please [register](#)
2. If you are not logged in, please [login](#)
3. Once logged in, copy the code displayed below to the file **\$HOME/.cdsapirc** (in your Unix/Linux environment)

```
url: https://cds.climate.copernicus.eu/api  
key: 99e5a58b-7eca-4419-98a4-74ba06c1f77a
```

2. Install the CDS API client

The CDS API client is a Python based library. It provides support for Python 3.

You can install the CDS API client via the package management system pip, by running on Unix/Linux the command below.

```
$ pip install "cdsapi>=0.7.7"
```

- ① Use of latest version available is highly recommended.

3. Use the CDS API client for data access

Once the CDS API client is installed, it can be used to request data from the datasets listed in the CDS, ADS, ECDS, XDS and CEMS Early Warning DS catalogues.

One must **agree to the Terms of Use** of a dataset before downloading any data out of it. This step must be done manually from the dataset page (at the bottom of the download form).

At the bottom of each dataset download form, press the "**Show API request code**" button to display a Python code snippet that can be used to run the manually built request. The API call must have this syntax:

```
import cdsapi

client = cdsapi.Client()

dataset = "<DATASET-SHORT-NAME>"
request = {
    <SELECTION-REQUEST>
}
target = "<TARGET-FILE>"

client.retrieve(dataset, request, target)
```

Example

```
import cdsapi

client = cdsapi.Client()

dataset = 'reanalysis-era5-pressure-levels'
```

```
request = {
    'product_type': ['reanalysis'],
    'variable': ['geopotential'],
    'year': ['2024'],
    'month': ['03'],
    'day': ['01'],
    'time': ['13:00'],
    'pressure_level': ['1000'],
    'data_format': 'grib',
}
target = 'download.grib'

client.retrieve(dataset, request, target)
```

New API package for advanced users

A new API package — [ecmwf-datastores-client](#) — is now available, with advanced features: metadata retrieval, synchronous or asynchronous job submission, REST API.

For setup instructions, see the [documentation](#). Technical details are available in the [Retrieve API](#), [Catalogue API](#) and [User profile API](#) documentation. Please note that REST API is recommended for advanced users and is not supported.

Also provided are [Jupyter Notebooks Tutorials](#) which include code in Python and content in Markdown to provide clear, engaging and practical instructions on using some of the most popular ecmwf-datastores-client functions.

[ecmwf-datastores-client](#), part of the ECMWF earthkit.data package, is **Incubating** and subject to ECMWF's guidelines on [Software Maturity](#). This means that the new API package is mostly feature-complete and the interface is mostly stable. If this software is to be used in operational systems you are strongly advised to use a released tag in your system configuration, and **you should be willing to accept incoming changes and bug fixes that require adaptations on your part**.

Users are not requested to migrate to using the new API package at this time and may continue to use the existing [cdsapi](#) package until further

notice.

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[CEMS Early Warning Data Store](#)

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